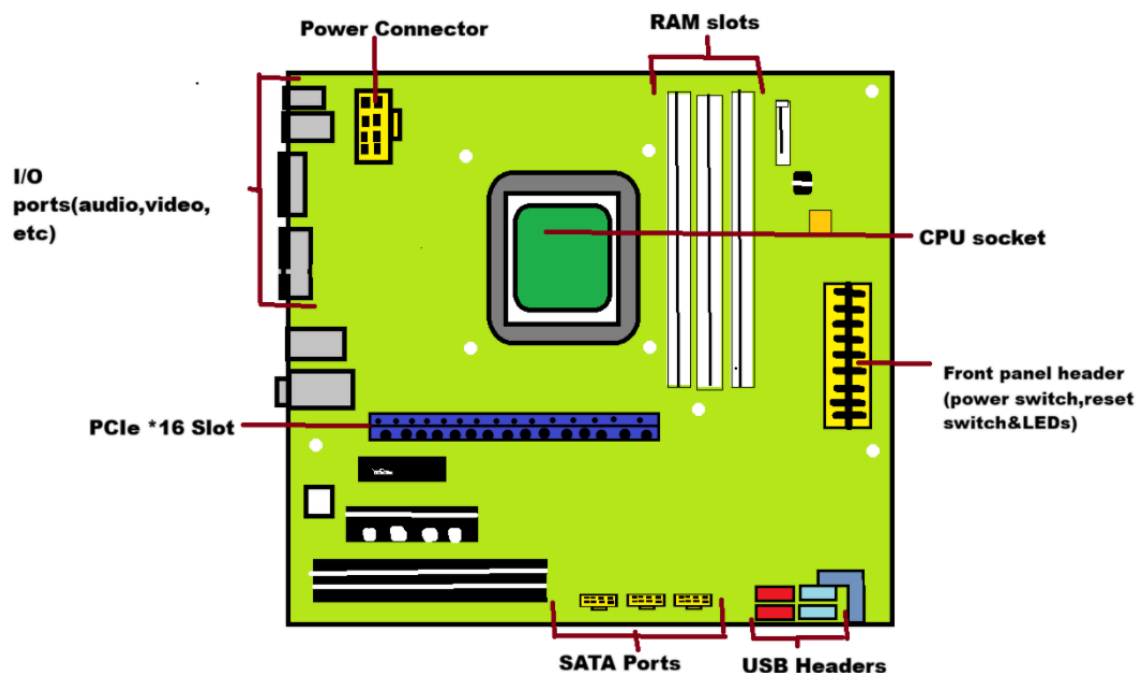


Assignment-1.2

“Motherboard and CPU”

TASK-1

Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



- **CPU Socket** – The CPU is installed here.
- **RAM Slots** – Used to install RAM.
- **PCIe x16 Slot** – Usually used for a graphics card.
- **SATA Ports** – Used to connect SATA HDDs or SSDs.
- **CPU Power Connector** – Provides power to the CPU.
- **USB Header** – Connects USB ports on the computer case.
- **I/O ports-** Used to connect ports like audio,video,etc.
- **Front panel Header-** used to connect switch, reset,LEDs.

TASK-2

Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

CPU Socket	Compatible CPU
LGA1151	Intel Core i7-9700K
AM4	AMD Ryzen 5 5600
LGA1200	Intel Core i5-10600K

- **LGA1151(Land grid array)-** Lga has flat electrical points, while there are tiny pins on socket. **1151** means the number of connections are present o LGA. So Intel designed a socket with **1,151 contacts** for that platform.
- **AM4-** . AM is AMD socket designation. AM4 uses **PGA(pin grind array)**. In a traditional PGA arrangement, the **CPU has pins**, while the motherboard socket has holes that receive them. AM\$ supports AMD Ryzen 5 5600 CPU.
- **LGA1200-** It is also a land grid array type socket. But it has 1200 flat electric points for connections. LGA supports Intel Core i5-10600K .

TASK -3 Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

- ☐ **CPU-** It is a processing unit, CPU processes instruction and does calculations.
- ☐ **Chipset-** It is used to manage communication between CPU and various Motherboard components(storage, USB devices, audio controllers, network chips) or features efficiently at the same time and speed.

>>What happens when we try to open SPOTIFY????

Step 1:

The user clicks the icon, and the Operating System sends a command to launch Spotify.

Step 2:

The OS locates the Spotify program on the **SSD/Storage** drive.

Step 3:

The motherboard's **chipset** helps transfer the Spotify app data from the storage drive directly into **RAM** for fast access.

Step 4:

Spotify connects to the internet via the network controller to fetch user data, track information, and lyrics.

Step 5:

The **CPU** executes the program instructions, processes the incoming audio stream from the internet, and coordinates with the hardware.

Step 6:

The audio hardware/sound controller processes the digital data into analog sound signals.

Step 7:

The final audio signal is sent through the computer's audio jack to your headphones or speakers.

TASK -4 Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.

